CLASS 10 MH BOARD

SCIENCE 1

CHAPTERWISE NOTES

Chapter 1: Gravitation

Introduction

- The chapter covers fundamental concepts of gravitation, planetary motion, and their implications.
- Gravitational force is a universal force acting between any two objects in the universe.

Key Concepts

- **Gravitation:** Discovered by Sir Isaac Newton after observing an apple fall from a tree. He concluded that the Earth attracts the apple towards its center.
- **Centripetal Force:** A force that acts on any object moving in a circle, directed towards the center of the circle. It is essential for circular motion; without it, an object would fly off in a straight line.
- **Kepler's Laws:** Three laws describing planetary motion, formulated by Johannes Kepler based on astronomical data.
    - **First Law:** The orbit of a planet is an ellipse with the Sun at one of the foci.
    - **Second Law:** The line joining the planet and the Sun sweeps equal areas in equal intervals of time.
    - **Third Law:** The square of a planet's period of revolution (T^2) is directly proportional to the cube of its mean distance from the Sun (r^3).
        - **Formula:** T^2/r^3 = {constant} = K

Newton's Universal Law of Gravitation

- Every object in the universe attracts every other object with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- **Formula:** $F = G(m1\ m2)/d^2$
    - **G:** Universal Gravitational Constant.
    - **Value of G:** 6.673×10^{-11} $Nm^2\ kg^{-2}$ (First measured by Henry Cavendish).
    - **Unit of G:** $N\ m^2\ kg^{-2}$

Earth's Gravitational Acceleration (g)

- The gravitational force exerted by the Earth causes an object to accelerate. This is called acceleration due to gravity (g).
- The value of g on the Earth's surface is approximately $9.77\ m/s^2$.
- **Formula:** $g = GM/R^2$
    - **M:** Mass of the Earth (6×10^{24} kg).
    - **R:** Radius of the Earth (6.4×10^{6} m).

Variation in the Value of g

- **Change along the surface:** The Earth is not a perfect sphere; it bulges at the equator and is flatter at the poles.
  - g is highest at the poles (9.832 m/s^2) and lowest at the equator (9.78 m/s^2).
- **Change with height:** The value of g decreases as you move away from the Earth's surface.
- **Change with depth:** The value of g also decreases as you go deeper inside the Earth.
  - At the center of the Earth, the value of g is zero.

Mass and Weight

Feature	**Mass**	**Weight**
Definition	Amount of matter in an object.	The force with which the Earth attracts an object.
Quantity	Scalar quantity.	Vector quantity.
Unit	kilogram (kg).	Newton (N).
Consistency	Value is the same everywhere.	Changes from place to place.
Formula	-	W = F = mg

Free Fall and Escape Velocity

- **Free Fall:** When an object moves under the influence of gravity alone.
  - In a true free fall (in vacuum), all objects fall at the same rate, regardless of mass.
- **Escape Velocity:** The minimum initial velocity required for an object to escape the gravitational force of a planet forever.
  - **Formula:** v_{esc} = sqrt{(2GM)/R}
  - **Earth's Escape Velocity:** 11.2 km/s

Chapter 2: Periodic Classification of Elements

Early Attempts at Classification

- **Döbereiner's Triads (1817):**
    - Proposed that properties of elements are related to their atomic masses.
    - Arranged elements in groups of three (triads) with similar chemical properties.
    - The atomic mass of the middle element was approximately the mean of the other two.
- **Newlands' Law of Octaves (1866):**
    - Arranged elements in increasing order of atomic masses.
    - Found that every eighth element had properties similar to the first.
    - This law was only applicable up to the element Calcium.

Mendeleev's Periodic Table (1869-1872)

- Developed by Russian scientist Dmitri Mendeleev.
- **Periodic Law:** Properties of elements are a periodic function of their atomic masses.
- **Structure:**
    - **Groups:** Vertical columns.
    - **Periods:** Horizontal rows.
- **Merits:**
    - Predicted the existence and properties of undiscovered elements (e.g., Eka-Boron, Eka-Aluminium, Eka-Silicon).
    - Corrected the atomic masses of some elements.
    - Accommodated noble gases later on by creating a 'zero' group.
- **Demerits:**
    - Ambiguity in the position of Cobalt and Nickel due to similar atomic masses.
    - Position of the isotopes was not clear.
    - The position of hydrogen was not fixed.

Modern Periodic Table

- **Modern Periodic Law:** Properties of elements are a periodic function of their atomic numbers.
- **Structure:**
    - **Periods:** 7 horizontal rows.
    - **Groups:** 18 vertical columns.
    - **Blocks:** S-block (groups 1, 2), p-block (groups 13-18), d-block (groups 3-12), and f-block (lanthanides and actinides).
    - **Metalloids:** Lie along a zig-zag line in the p-block, separating metals (left side)

from nonmetals (right side).

Periodic Trends

- **Valency:**
    - Remains the same down a group.
    - Varies across a period.
- **Atomic Size:**
    - **Across a period (left to right):** Decreases due to increasing nuclear charge pulling electrons closer.
    - **Down a group:** Increases because new shells are added, increasing the distance between the nucleus and outermost electrons.
- **Metallic-Nonmetallic Character:**
    - **Metallic Character (Electropositivity):** The tendency to lose electrons. Increases down a group and decreases across a period.
    - **Nonmetallic Character (Electronegativity):** The tendency to gain electrons. Increases across a period and decreases down a group.

Chapter 3: Chemical Reactions and Equations

Chemical Reactions

- A process where old chemical bonds break and new bonds form, transforming reactants into new products.
- **Reactants:** Substances taking part in a reaction.
- **Products:** New substances formed as a result.
- **Chemical Equation:** A representation of a chemical reaction using chemical formulas.

Balancing Chemical Equations

- According to the Law of Conservation of Mass, the total mass of reactants must equal the total mass of products.
- An equation is balanced when the number of atoms of each element is the same on both sides.
- Balancing is done step-by-step using a trial and error method, adjusting coefficients.

Types of Chemical Reactions

- **Combination Reaction:** Two or more reactants combine to form a single product.
  - Example: $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$
- **Decomposition Reaction:** A single reactant breaks down into two or more products.
  - Example: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
- **Displacement Reaction:** A more reactive element displaces a less reactive element from its compound.
  - Example: $CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$
- **Double Displacement Reaction:** Ions are exchanged between reactants to form a precipitate.
  - Example: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
- **Endothermic vs. Exothermic:**
  - **Exothermic:** Reactions that release heat. (e.g., $CaO+H_2O \rightarrow Ca(OH)_2 + heat$)
  - **Endothermic:** Reactions that absorb heat. (e.g., $CaCO_3+heat \rightarrow CaO+CO_2$)

Factors Affecting the Rate of Reaction

- **Nature of Reactants:** More reactive reactants react faster.
- **Size of Particles:** Smaller particle size increases the reaction rate.

- **Concentration of Reactants:** Higher concentration increases the reaction rate.
- **Temperature of Reaction:** Higher temperature increases the reaction rate.
- **Catalyst:** A substance that increases the reaction rate without being consumed.

Oxidation and Reduction (Redox Reactions)

- **Oxidation:** A reaction where a reactant combines with oxygen, loses hydrogen, or loses one or more electrons.
- **Reduction:** A reaction where a reactant gains hydrogen, loses oxygen, or gains one or more electrons.
- **Redox Reaction:** A reaction where oxidation and reduction occur simultaneously.
- **Corrosion:** The process of gradual damage to a metal due to oxidation by atmospheric components.
    - **Rusting:** Corrosion of iron, forming a reddish layer of hydrated iron oxide ($Fe_2O_3 \cdot xH_2O$).
- **Rancidity:** The foul odor and taste developed in old oils and fatty foods due to air oxidation.

Chapter 4: Effects of Electric Current

Energy Transfer in an Electric Circuit

- A cell provides energy to a charge (Q) to flow through a circuit.
- This energy is converted into heat in a resistor, increasing its temperature.
- **Electrical Power:** $P = V \times I$
- **Heat Produced:** $H = P \times t = V \times I \times t$
    - **Joule's Law of Heating:** $H = I^2 \times R \times t$

Heating Effect of Electric Current

- When a current flows through a resistor, heat is produced. This is the heating effect of current.
- This principle is used in devices like electric cookers, heaters, and bulbs.
- **Fuse Wire:** A safety device that melts and breaks a circuit if an excessive current flows, preventing damage.
- **Short Circuit:** Occurs when live and neutral wires touch, causing a very large current and producing intense heat, which can lead to fires.

Magnetic Effect of Electric Current

- A magnetic field is produced around a current-carrying conductor.
- **Right-Hand Thumb Rule:** If you hold a conductor with your right hand with your thumb pointing in the direction of current, your fingers wrapping around the wire show the direction of the magnetic field lines.
- **Solenoid:** A coil of wire that produces a uniform magnetic field similar to a bar magnet when current flows through it.

Force on a Current-Carrying Conductor

- A force is exerted on a current-carrying conductor placed in a magnetic field.
- **Fleming's Left-Hand Rule:** Used to determine the direction of this force.
    - Thumb: Direction of Force.
    - Index Finger: Direction of Magnetic Field.
    - Middle Finger: Direction of Current.
- **Electric Motor:** A device that converts electrical energy into mechanical energy. It works on the principle of a force being exerted on a current-carrying coil in a magnetic field.

Electromagnetic Induction

- An electric current can be produced in a conductor by a changing magnetic field.
- **Faraday's Law of Induction:** A current is induced in a coil whenever the number of magnetic lines of force passing through it changes. This is called induced current.
- **Fleming's Right-Hand Rule:** Used to find the direction of the induced current.
    - Thumb: Direction of Motion.
    - Index Finger: Direction of Magnetic Field.
    - Middle Finger: Direction of Induced Current.
- **Electric Generator:** A device that converts mechanical energy into electrical energy based on electromagnetic induction.

Chapter 5: Heat

Latent Heat

- **Latent Heat of Fusion:** Heat energy absorbed by a substance at a constant temperature to change from a solid to a liquid state.
- **Latent Heat of Vaporization:** Heat energy absorbed by a substance at a constant temperature to change from a liquid to a gaseous state
- The temperature remains constant during a phase change because the absorbed heat is used to weaken molecular bonds.

Anomalous Behavior of Water

- Unlike other liquids, water contracts when heated from 0^0C to 4^0C.
- Its volume is minimum and its density is maximum at 4^0C.
- This behavior is crucial for aquatic life in cold regions, as water at 4^0C settles at the bottom of lakes, preventing them from freezing completely.

Dew Point and Humidity

- **Humidity:** The presence of water vapor (moisture) in the atmosphere.
- **Saturated Air:** Air that contains the maximum possible amount of water vapor at a given temperature.
- **Dew Point Temperature:** The temperature at which unsaturated air becomes saturated with water vapor.
- **Relative Humidity:** The ratio of the actual mass of water vapor in the air to the mass of vapor needed for saturation at the same temperature.
  - **Formula:** (Actual mass of water vapor)/ (Mass of vapor for saturation) ×100

Specific Heat Capacity

- **Definition:** The amount of heat energy required to raise the temperature of a unit mass of an object by 1^0C.
- **Unit:** J/kg·°C or cal/g·°C
- **Formula:** Heat absorbed or lost = Heat $(Q)=m\times c\times\Delta T$
- **Principle of Heat Exchange:** In an isolated system, the heat energy lost by a hot object equals the heat energy gained by a cold object.

Chapter 6: Refraction of Light

Refraction of Light

- **Definition:** The change in the direction of a light ray when it passes from one transparent medium to another.
- **Causes:** The velocity of light changes as it moves from one medium to another.
- **Key terms:**
  - **Incident Ray:** The incoming ray.
  - **Refracted Ray:** The ray that changes direction inside the new medium.
  - **Normal:** A line perpendicular to the surface at the point of incidence.
  - **Angle of Incidence (i):** Angle between the incident ray and the normal.
  - **Angle of Refraction (r):** Angle between the refracted ray and the normal.

Laws of Refraction

1. The incident ray, the refracted ray, and the normal all lie in the same plane.
2. For a given pair of media, the ratio of \sin(i) to \sin(r) is constant. This is known as **Snell's Law.**
   - **Formula:** $\sin i / \sin r$ = constant = n

Refractive Index (n)

- A measure of how much a light ray changes direction when entering a medium.
- **Formula:** ${}^1n_2 = v_1 / v_2$
  - v1: Velocity of light in medium 1.
  - v2: Velocity of light in medium 2.
- **Absolute Refractive Index:** Refractive index with respect to a vacuum.
- **Behavior of light:**
  - **Rarer to Denser Medium:** Light bends **towards** the normal.
  - **Denser to Rarer Medium:** Light bends **away** from the normal.

Atmospheric Refraction

- **Twinkling of Stars:** The changing refractive index of the atmosphere due to air motion and temperature variations causes star light to bend constantly, making them appear to twinkle.
- **Advanced Sunrise and Delayed Sunset:** We see the Sun slightly before it rises and after it sets because atmospheric refraction bends the light from the Sun towards us.

Dispersion and Total Internal Reflection

- **Dispersion of Light:** The separation of white light into its component colors (VIBGYOR) when passing through a medium like a prism.
    - Red light bends the least, and violet light bends the most.
- **Total Internal Reflection (TIR):** When light traveling in a denser medium hits the boundary with a rarer medium at an angle greater than the critical angle, all the light is reflected back into the denser medium.
- **Rainbow:** A natural phenomenon caused by the combination of dispersion, refraction, and total internal reflection of sunlight in water droplets.

Chapter 7: Lenses

Introduction to Lenses

- A lens is a transparent medium with two spherical surfaces.
- **Convex (Converging) Lens:** Thicker at the center, converges parallel light rays to a point (the principal focus).
- **Concave (Diverging) Lens:** Thinner at the center, diverges parallel light rays as if they are coming from a point (the principal focus).
- **Key terms:**
  - **Optical Centre (O):** A point on the principal axis where light rays pass without changing their path.
  - **Principal Focus (F):** The point on the principal axis where parallel rays converge (convex) or appear to diverge from (concave) after refraction.
  - **Focal Length (f):** The distance between the optical center and the principal focus.

Image Formation by Lenses

- **Rules for Ray Diagrams:**
  1. Parallel incident ray passes through the focus (convex) or appears to come from the focus (concave).
  2. Ray passing through the focus becomes parallel after refraction.
  3. Ray passing through the optical center goes straight.
- **Convex Lens Images:** Can form real, inverted, and virtual, erect images depending on the object's position.
- **Concave Lens Images:** Always forms virtual, erect, and smaller images.

Lens Formula and Power of a Lens

- **Lens Formula:** Relates object distance (u), image distance (v), and focal length (f).
  - **Formula:** $1/v - 1/u = 1/f$
- **Power of a Lens (P):** The ability of a lens to converge or diverge light rays.
  - **Formula:** $P = 1/f$ (in meters).
  - **Unit:** Dioptre (D).

Human Eye and Vision Defects

- **Human Eye Structure:**
  - **Cornea:** Transparent outer cover where light first refracts.
  - **Iris:** Dark, fleshy screen controlling pupil size.
  - **Pupil:** Hole in the iris that adjusts to light intensity.
  - **Crystalline Lens:** Double convex lens that adjusts focal length to focus images.

    - **Retina:** Light-sensitive screen where the real, inverted image is formed.
- **Power of Accommodation:** The ability of the eye lens to adjust its focal length to see objects at different distances.
- **Nearsightedness (Myopia):** Nearby objects are clear, but distant objects are indistinct.
    - **Cause:** Lens has a high converging power or the eyeball is too long.
    - **Correction:** Concave lens.
- **Farsightedness (Hypermetropia):** Distant objects are clear, but nearby objects are indistinct.
    - **Cause:** Lens has a low converging power or the eyeball is too short.
    - **Correction:** Convex lens.
- **Presbyopia:** Loss of accommodation power with age.
    - **Correction:** Bifocal lenses.

Chapter 8: Metallurgy

Physical Properties

- **Metals:**
  - Mostly solids at room temperature (exceptions: mercury, gallium).
  - Have metallic luster, are ductile, and malleable.
  - Good conductors of heat and electricity.
  - Generally hard (exceptions: alkali metals like Na, K).
  - High melting and boiling points (exception: Na, K, Hg, Ga).
- **Nonmetals:**
  - Can be solids, liquids (exception: bromine), or gases.
  - No luster (exception: iodine).
  - Not hard (exception: diamond).
  - Low melting and boiling points.
  - Bad conductors of heat and electricity (exception: graphite).

Chemical Properties of Metals

- **Reaction with Oxygen:** Metals form metal oxides on heating in air. Highly reactive metals like Na and K react at room temperature.
- **Reaction with Water:**
  - Highly reactive metals (Na, K) react vigorously, producing H_2 gas and heat.
  - Less reactive metals (Al, Fe, Zn) react only with steam.
- **Reactivity Series:** An arrangement of metals in increasing or decreasing order of their reactivity.
  - **Highly reactive:** K, Na, Ca, Mg, Al
  - **Moderately reactive:** Zn, Fe, Sn, Pb
  - **Less reactive:** Cu, Hg, Ag, Au

Metallurgy: Extraction of Metals

- **Ore:** A mineral from which a metal can be extracted economically.
- **Gangue:** Impurities like soil and sand present in ores.
- **Steps of Extraction:**
  1. **Concentration of Ores:** Removing gangue. Methods include separation based on gravitation (Wilfley table, hydraulic separation), magnetic separation, and froth floatation.
  2. **Extraction of Metals:**
     - **Reactive Metals:** Electrolytic reduction of their molten salts. (e.g., Na from molten NaCl).
     - **Moderately Reactive Metals:** Converted to oxides by roasting (for sulfide ores) or calcination (for carbonate ores), then reduced with a suitable

reductant like carbon.

- **Less Reactive Metals:** Found in free state or extracted by heating their ores in air.

3. **Refining of Metals:** Purification of crude metals, usually by electrolysis.

Corrosion and its Prevention

- **Corrosion:** The damage of a metal due to oxidation by atmospheric components.
  - Examples: Rusting of iron, green layer on copper, black layer on silver.
- **Prevention Methods:**
  - Applying a protective layer of paint, oil, or grease.
  - **Galvanizing:** Coating with a layer of zinc.
  - **Electroplating:** Coating with a less reactive metal using electrolysis.
  - **Alloying:** Creating a homogeneous mixture of metals or nonmetals to enhance properties and decrease corrosion.

Chapter 9: Carbon Compounds

Bonds in Carbon Compounds

- Carbon compounds generally have low melting and boiling points and are poor conductors of electricity.
- This is because carbon atoms form **covalent bonds** by sharing electrons, rather than ionic bonds.
- **Electron-dot structure:** A diagram showing shared valence electrons between atoms.

Carbon: A Versatile Element

- The unique properties of carbon lead to an extremely large number of compounds.
- **Catenation:** The ability of carbon atoms to form strong covalent bonds with other carbon atoms, creating long chains or rings.
- **Tetravalency:** Carbon has a valency of four, allowing it to form bonds with four other atoms.
- **Multiple Bonds:** Carbon can form single, double, or triple bonds, increasing the variety of compounds.

Hydrocarbons

- Compounds containing only carbon and hydrogen.
- **Saturated Hydrocarbons (Alkanes):** Contain only single bonds between carbon atoms.
- **Unsaturated Hydrocarbons (Alkenes & Alkynes):** Contain double or triple bonds between carbon atoms.
    - Alkenes: C=C double bonds.
    - Alkynes: C≡C triple bonds.
- **Isomerism:** Compounds with the same molecular formula but different structural formulas.
- **Homologous Series:** A series of compounds with the same functional group but varying chain length, where neighboring members differ by one -CH_2- unit.

Functional Groups

- A heteroatom (e.g., O, N, S, halogens) or group of atoms attached to a carbon chain that gives the compound its specific chemical properties.
- **Nomenclature:**
    - **IUPAC System:** A logical, worldwide-accepted naming system based on the structure of compounds.
    - Names are formed from a parent name (based on the carbon chain length) and a prefix/suffix (based on functional groups).

Chemical Properties of Carbon Compounds

- **Combustion:** Carbon compounds burn in oxygen to produce CO_2, water, heat, and light.
- **Oxidation:** Addition of oxygen or removal of hydrogen from a compound, often facilitated by oxidizing agents like potassium permanganate.
- **Addition Reaction:** Unsaturated compounds with multiple bonds react with other compounds to form a single, saturated product.
- **Substitution Reaction:** An atom or group of atoms is replaced by another. Saturated hydrocarbons undergo this reaction, often in the presence of sunlight.

Chapter 10: Space Missions

Introduction to Space Missions

- Space missions involve launching artificial satellites into Earth's orbit or sending spacecraft to explore outer space.
- They are crucial for communication, weather forecasting, military surveillance, and scientific research.
- **Natural Satellite:** An astronomical object orbiting a planet (e.g., the Moon).
- **Artificial Satellite:** A man-made object orbiting Earth or another planet.

Orbits of Artificial Satellites

- The function of a satellite determines its orbit's height, shape (circular/elliptical), and inclination.
- **Critical Velocity (v_c):** The specific tangential velocity required for a satellite to revolve in a stable orbit.
  - **Formula:** $v_c = \sqrt{GM/(R+h)}$
- **Orbit Classification by Height:**
  - **High Earth Orbits (HEO):** Height > 35,780 km. Satellites here take ~24 hours to orbit, making them appear stationary relative to Earth (**geosynchronous satellites**).
  - **Medium Earth Orbits (MEO):** Height between 2,000 km and 35,780 km. Used by navigation satellites (e.g., GPS).
  - **Low Earth Orbits (LEO):** Height between 180 km and 2,000 km. Used for scientific experiments and by the International Space Station.

Satellite Launch Vehicles

- Used to place satellites into their specific orbits.
- Work based on **Newton's Third Law of Motion**.
- Launch vehicles are often multi-staged to reduce weight as fuel is consumed.
- **ISRO's Launchers:** Polar Satellite Launch Vehicle (PSLV) and Geosynchronous Satellite Launch Vehicle (GSLV).

Space Missions to Other Objects

- **Moon Missions:** The first missions to explore another celestial body. The Chandrayaan-1 mission by ISRO discovered the presence of water on the moon.
- **Mars Missions:** ISRO's Mangalyaan mission successfully orbited Mars, providing valuable information about its surface and atmosphere.
- **Escape Velocity:** The minimum velocity an object needs to overcome a planet's gravity.

    - **Earth's Escape Velocity:** 11.2 km/s
- **Space Debris:** Non-functional satellites and other man-made objects orbiting Earth, posing a collision risk to new missions. Management of this debris is a growing concern.

THANK YOU

SVK STUDY